

Scalability & Future Plan

Date	5 july 2026
Team ID	XXXXXX
Project Name	Opti crop- Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Monolithic Application Architecture	Both the Flask frontend UI and the ML execution backend run as a single process, making it difficult to update the model without taking the entire website offline.	High
2	Lack of Input Sanitization & Data Validation	The application processes raw text fields directly into NumPy arrays without defensive checks, risking application crashes if anomalous or empty data is submitted	High
3	Single-Node Local Processing	The ML model relies completely on local system memory for inference computations, leading to request timeouts under simultaneous testing.	medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Tied to a localized development runtime environment.	Deploy the system inside Docker containers orchestrated by Kubernetes (K8s) to automatically scale application instances based on incoming traffic volume.
2	Data Storage	Local flat files handle agricultural record parameters.	Migrate to a cloud-managed NoSQL environment like MongoDB for flexible, schema-less logging of unstructured environmental inputs.
3	Performance	Synchronous requests experience latency bottlenecks during model execution.	Decouple the ML engine using FastAPI as a dedicated inference microservice, introducing an Amazon S3 bucket to store and dynamically pull updated model binaries (.pkl).

4	Security	Unauthenticated user endpoints with no data encryption.	Implement OAuth 2.0 / JWT Authentication tokens to secure API endpoints, alongside strict data parsing libraries (like Pydantic) to clean user inputs.
---	----------	---	---

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
-------	-------------------------------	-----------------	-----------------

Phase 2	Localized Weather API Integration: Connect live geolocation coordinates to third-party weather tracking services	3 months	Eliminates the need for manual rainfall and temperature entries by automatically fetching real-time environmental data.
Phase 3	Multi-Language Support (Localization): Translate the interface and crop analysis outputs into regional vernacular languages.	6 months	Lowers the technical barrier to entry, allowing non-English speaking regional farmers to seamlessly use the engine.
Phase 4	Deep Learning Yield Prediction: Incorporate Artificial Neural Networks (ANNs) to predict actual yield weights per acre	12 months	Enhances the application from a baseline crop selection tool into a robust agricultural financial planning and forecasting asset.